Thrinayani Yedhoti

Redmond, WA | +1 (425) 321-9128 | thri9e@uw.edu | linkedin.com/in/thrinayani-yedhoti | github.com/Thrinayani39e | F-1 CPT/OPT Eligible

Summary

MS Computer Science candidate at University of Washington (GPA: 3.83) with 1.5+ years of backend and full-stack engineering experience shipping production software at Schneider Electric R&D (industrial automation). Built REST APIs, backend services, dashboards, and access-control systems for manufacturing plant operations. Seeking Summer/Fall 2026 Software Engineer Internships.

Skills

Languages: Java, Python, TypeScript, C#, C++, JavaScript, SQL
Backend & APIs: FastAPI, Spring Boot, ASP.NET, Node.js, REST APIs, .NET, PostgreSQL, MySQL, Redis, SQLite, SQLAlchemy (ORM)
Dashboards & Frontend: Angular, WPF, ASP.NET, RBAC, OAuth
Data & Automation: Real-time edge computing, CI/CD, Docker, Docker Compose, Azure DevOps
AI/ML: PyTorch, CNNs, Spiking Neural Networks, Computer Vision, OpenCV, LLM Integration (Copilot, Claude)
Methods: Agile/SCRUM, Unit Testing, Code Review, System Design, Large Codebase Navigation, SonarQube, Coverity

Education

MS, Computer Science and Software Engineering — **Expected: March 2027**
University of Washington, Bothell — *GPA: 3.83/4.0*

- **Coursework:** Advanced Computer Vision, Algorithm Analysis & Design, Research Methods in SE, Evaluating Software Design, Software Architecture, Generative AI | **CS Grader:** Introduction to AI (Jan - Mar 2026)

Experience

Schneider Electric – R&D, Industrial Automation — **Feb 2025 – Aug 2025**
Software Design Engineer — *Bengaluru, India*

- Reduced PLC license validation time by 30% by engineering a GSE mechanism for Unity M580 industrial controllers using **C#** and **WPF** with full unit test coverage, directly supporting factory automation workflows.
- Decreased carbon emission reporting latency for manufacturing plants by architecting a real-time **edge computing pipeline** using **Python** and **FastAPI**, improving operational visibility into production metrics.
- Accelerated vulnerability inspection workflows by building **user management dashboards** and RBAC-based access controls using **Angular** and **ASP.NET**.

Schneider Electric – R&D, Industrial Automation — **Aug 2024 – Feb 2025**
Graduate Engineer Trainee — *Bengaluru, India*

- Reduced application load times by 20% by refactoring **.NET**, **WPF**, and **Angular** codebases supporting industrial automation software, with maintained unit test coverage across all modules.
- Accelerated database query execution by 40% by designing an **ORM-based data architecture** using **SQLite** and **SQLAlchemy**.

Schneider Electric – R&D, Industrial Automation — **Jan 2024 – Jul 2024**
Software Engineer Intern — *Bengaluru, India*

- Shipped production-ready **RBAC system** using **C#**, **Angular**, and **.NET** with zero vulnerabilities across Sonar and Coverity static analysis; delivered clean, well-tested code reviewed by senior engineers.

Projects

Distributed Rate Limiter Microservice (Java, Spring Boot, Redis, PostgreSQL, Docker) | **Link** — **May 2026**

- Built production-ready distributed rate-limiting **backend service** in **Java** and **Spring Boot 3** implementing token bucket and sliding window algorithms via **Redis**; deployed live via **Docker Swarm** with **CI/CD** through GitHub Actions, demonstrating end-to-end ownership of a real production system.
- Persisted full audit trail to **PostgreSQL** using **Hibernate/JPA**; exposed 3 **REST API** endpoints returning HTTP 429 on threshold breach with per-client stats, providing operational visibility into API usage and capacity.

Optimal Seamlines for Image Stitching via Fast Marching Method (C++, OpenCV) | **Link** — **Nov–Dec 2025**

- Outperformed GraphCutSeamFinder and DPSeamFinder in suppressing parallax artifacts by implementing an FMM-based seam finder in an **OpenCV** pipeline; collaborated cross-functionally with Lawrence Sanchez (Engineering Director, Minecraft/Microsoft).
- Solved Eikonal equation over fused cost fields (color, edge, texture, saliency) to compute globally optimal seam paths, delivering stable runtime and measurable quality improvements across benchmark datasets.

Publications

- Attention-Based Spiking Neural Networks for Fashion Product Labelling — *IEEE, 2024*
- CNN-Based Age Estimation using Diverse Facial Datasets — *IEEE, 2024*